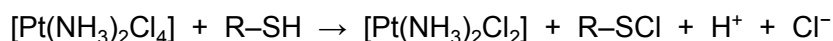
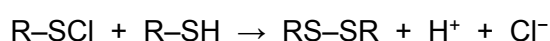


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The Pt(IV) complex *cis*-[Pt(NH₃)₂Cl₄] has been found to act as a prodrug by reacting with the S–H bond of thiols in the cytosol of a cell to produce the Pt-based antineoplastic chemotherapy drug cisplatin (*cis*-[Pt(NH₃)₂Cl₂]) and a disulfide oxidation product. Researchers studying the reaction between *cis*-[Pt(NH₃)₂Cl₄] and the thiolactate anion (R–SH, where R = –CH(CH₃)CO₂[–]) as a model thiol compound proposed that the reaction proceeds in two steps according to the mechanism shown by Reactions 1 and 2.



Reaction 1 (Step 1)



Reaction 2 (Step 2)

In the first step of the mechanism, cisplatin is generated along with a transient chlorothiolate anion (R–SCI) intermediate. In the second step, the intermediate reacts with a second R–SH molecule to form the disulfide oxidation product (RS–SR). The researchers found that the activation energy for Step 1 was significantly greater than the activation energy for Step 2.

A kinetic analysis was then performed by measuring the initial rate of the reaction in six different 20.0-mL trial solutions containing different initial molar concentrations of the Pt(IV) complex and R–SH. All reactions were conducted at 25 °C in solutions with physiological pH of 7.3. The results of the analysis are given in Table 1.

Table 1 Kinetic Analysis for the Reaction of *cis*-[Pt(NH₃)₂Cl₄] and R–SH at 25 °C in Solutions with pH 7.3

Trial	[Pt(IV)]	[R–SH]	Initial rate (M/s)
1	3.0 × 10 ^{–4}	3.0 × 10 ^{–4}	1.6 × 10 ^{–4}
2	3.0 × 10 ^{–4}	2.5 × 10 ^{–4}	1.3 × 10 ^{–4}
3	3.0 × 10 ^{–4}	2.0 × 10 ^{–4}	1.1 × 10 ^{–4}
4	3.0 × 10 ^{–4}	1.5 × 10 ^{–4}	8.0 × 10 ^{–5}
5	3.0 × 10 ^{–4}	1.0 × 10 ^{–4}	5.4 × 10 ^{–5}
6	1.5 × 10 ^{–4}	2.0 × 10 ^{–4}	5.4 × 10 ^{–5}

For the overall reaction to convert R–SH to the disulfide oxidation product, the overall rate constant *k'* was found to vary with the pH of the solution. Measured values for *k'* at several pH values are listed in Table 2.

Table 2 Dependence of the Overall Rate Constant *k'* on the Reaction Solution pH

pH	<i>k'</i> (M ^{–1} s ^{–1})
2.58	8 × 10 ^{–1}
2.91	1.4 × 10 ⁰
4.07	3 × 10 ⁰
5.12	3 × 10 ¹
6.11	7 × 10 ²
8.55	2 × 10 ⁴
9.08	5 × 10 ⁴
9.94	1 × 10 ⁵

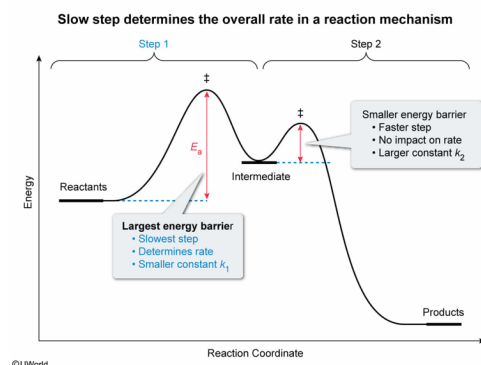
Based on the proposed mechanism in the passage, which of the following is true about the overall rate of the reaction to convert R–SH to the disulfide oxidation product?

- ✓ ☒ A. The rate-determining step of the overall reaction is Step 1 only. [87%]
☐ B. The rate-determining step of the overall reaction is Step 2 only. [5%]
☐ C. Both Step 1 and Step 2 determine the overall rate of the reaction. [6%]
☐ D. Neither Step 1 nor Step 2 determines the overall rate of the reaction.

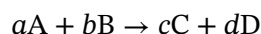
Correct

87% answered correctly

Explanation:



The general reaction in which reactants A and B form products C and D is given by:



The rate of this reaction can be expressed by the general rate law:

$$\text{Rate} = k[A]^m[B]^n$$

where k is the temperature-dependent rate constant, $[A]$ and $[B]$ are the molar concentrations of the reactants, and m and n are the partial **reaction orders** (determined experimentally). The temperature dependence of k is described by the **Arrhenius equation**:

$$k = Ae^{-\left(\frac{E_a}{RT}\right)}$$

The rate law shows that the **reaction rate** is **proportional to k** (ie, rate increases as k increases), and the

Arrhenius equation shows that k decreases exponentially as the **activation energy E_a** increases. If a reaction mechanism involves two or more steps, the overall reaction can proceed only as quickly as its **slowest step** (ie, the step with the **highest E_a** and **smallest k**). Consequently, the slow step is the **rate-determining step** of a reaction (ie, the slow step determines the rate law).

For the mechanism in Figure 1, the rate laws for each step are:

$$\text{Rate}_{(\text{Step 1})} = k_1[\text{Pt(IV)}][\text{R} - \text{SH}]$$

$$\text{Rate}_{(\text{Step 2})} = k_2[\text{R} - \text{SCl}][\text{R} - \text{SH}]$$

The passage states that the researchers found the activation energy of Step 1 to be significantly greater than the activation energy of Step 2. As such, the rate constant for Step 1 (k_1) is considerably less than the rate constant for Step 2 (k_2), and

$$\text{Rate}_{(\text{Step 1})} \ll \text{Rate}_{(\text{Step 2})}$$

Therefore, Step 1 is the rate-determining step of the overall reaction to convert R–SH to the disulfide oxidation product.

(Choices B, C, and D) Because the activation energy of Step 2 is smaller than that of Step 1, Step 2 is the fast step and is not rate-limiting for the overall reaction. Step 2 occurs quickly but cannot happen until Step 1 is completed. Consequently, Step 1 determines the overall rate of the reaction, and Step 2 makes no contribution to the overall rate.

Educational objective:

The rate of a reaction is proportional to the rate constant k , which decreases exponentially as the activation energy needed to reach the transition state increases. In a reaction involving two or more steps, the step that must overcome the largest energy barrier (the slowest step) is the rate-determining step.

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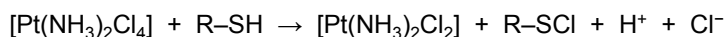
Subject:
General Chemistry

QID:403762

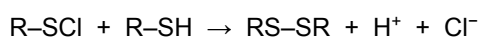
System:
Thermodynamics, Kinetics & Gas Laws

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The Pt(IV) complex $cis\text{-}[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$ has been found to act as a prodrug by reacting with the S–H bond of thiols in the cytosol of a cell to produce the Pt-based antineoplastic chemotherapy drug cisplatin ($cis\text{-}[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$) and a disulfide oxidation product. Researchers studying the reaction between $cis\text{-}[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$ and the thiolactate anion (R-SH , where $\text{R} = -\text{CH}(\text{CH}_3)\text{CO}_2^-$) as a model thiol compound proposed that the reaction proceeds in two steps according to the mechanism shown by Reactions 1 and 2.



Reaction 1 (Step 1)



Reaction 2 (Step 2)

In the first step of the mechanism, cisplatin is generated along with a transient chlorothiolate anion (R-SCI) intermediate. In the second step, the intermediate reacts with a second R-SH molecule to form the disulfide oxidation product (RS-SR). The researchers found that the activation energy for Step 1 was significantly greater than the activation energy for Step 2.

A kinetic analysis was then performed by measuring the initial rate of the reaction in six different 20.0-mL trial solutions containing different initial molar concentrations of the Pt(IV) complex and R-SH . All reactions were conducted at 25 °C in solutions with physiological pH of 7.3. The results of the analysis are given in Table 1.

Table 1 Kinetic Analysis for the Reaction of $cis\text{-}[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$ and R-SH at 25 °C in Solutions with pH 7.3

Trial	[Pt(IV)]	[R-SH]	Initial rate (M/s)
1	3.0×10^{-4}	3.0×10^{-4}	1.6×10^{-4}
2	3.0×10^{-4}	2.5×10^{-4}	1.3×10^{-4}
3	3.0×10^{-4}	2.0×10^{-4}	1.1×10^{-4}
4	3.0×10^{-4}	1.5×10^{-4}	8.0×10^{-5}
5	3.0×10^{-4}	1.0×10^{-4}	5.4×10^{-5}
6	1.5×10^{-4}	2.0×10^{-4}	5.4×10^{-5}

For the overall reaction to convert R-SH to the disulfide oxidation product, the overall rate constant k' was found to vary with the pH of the solution. Measured values for k' at several pH values are listed in Table 2.

Table 2 Dependence of the Overall Rate Constant k' on the Reaction Solution pH

pH	$k' \text{ (M}^{-1}\text{s}^{-1}\text{)}$
2.58	8×10^{-1}
2.91	4×10^0
4.07	3×10^0
5.12	3×10^1
6.11	7×10^2
8.55	2×10^4
9.08	5×10^4
9.94	1×10^5

Which of the following expressions gives the rate law for the overall reaction to convert R-SH to the disulfide oxidation product?

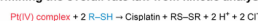
- ☐ A. Rate = $k[\text{R-SH}]$ [12%]
- ☒ B. Rate = $k[\text{Pt(IV)}][\text{R-SH}]$ [61%]
- ☐ C. Rate = $k[\text{Pt(IV)}][\text{R-SH}]^2$ [21%]
- ☐ D. Rate = $k[\text{R-SCI}]^2$ [4%]

Correct

62% answered correctly

Explanation:

Determining the overall rate law from kinetic analysis data



Trial	[Pt(IV)]	[R-SH]	Initial rate (M/s)
1	3.0×10^{-4}	3.0×10^{-4}	1.6×10^{-4}
2	3.0×10^{-4}	2.5×10^{-4}	1.3×10^{-4}
3	3.0×10^{-4}	2.0×10^{-4}	1.1×10^{-4}
4	3.0×10^{-4}	1.5×10^{-4}	8.0×10^{-5}
5	3.0×10^{-4}	1.0×10^{-4}	5.4×10^{-5}
6	1.5×10^{-4}	2.0×10^{-4}	5.4×10^{-5}

$$\text{Rate} = k[\text{Pt(IV)}][\text{R-SH}]$$

First-order dependence on [R-SH] $\rightarrow n = 1$

First-order dependence on [Pt(IV)] $\rightarrow m = 1$

A **rate law** relates the rate of a reaction to the **molar concentration** of each reactant. A rate law is expressed generally by the equation:

$$\text{Rate} = k[\text{A}]^m[\text{B}]^n$$

where k is a constant and m and n are values indicating the dependence of the rate on the respective concentrations of reactants A and B. For reactions involving two or more steps, the values of m and n must be **determined experimentally** from kinetics measurements.

For the reaction given in the passage, substituting the Pt(IV) complex and R-SH reactants into the general rate law gives:

$$\text{Rate} = k'[\text{Pt(IV)}]^m[\text{R-SH}]^n$$

The values of m and n can be evaluated from the reactant concentrations and initial rates given in Table 1 by setting up a **proportion of rates** (several combinations are possible). For example, applying a proportion of Trial 3 to Trial 6 (where [Pt(IV)] changes by a factor of 2 but [R-SH] remains constant) gives:

$$\frac{\text{Rate}_3}{\text{Rate}_6} = \frac{k'[\text{Pt(IV)}]_3^m[\text{R-SH}]_3^n}{k'[\text{Pt(IV)}]_6^m[\text{R-SH}]_6^n}$$

Substituting the given values from Table 1 (with units canceled) yields:

$$\frac{1.1 \times 10^{-4}}{5.4 \times 10^{-5}} = \frac{k' \cdot (3.0 \times 10^{-4})^m \cdot (2.0 \times 10^{-4})^n}{k' \cdot (1.5 \times 10^{-4})^m \cdot (2.0 \times 10^{-4})^n}$$

Canceling common terms and solving for m gives:

$$\begin{aligned} 2.0 &= \frac{(3.0 \times 10^{-4})^m}{(1.5 \times 10^{-4})^m} \\ 2.0 &= (2.0)^m \\ m &= 1 \end{aligned}$$

Similarly, applying a proportion of Trial 1 to Trial 4 (where [Pt(IV)] remains constant but [R-SH] changes by a factor of 2) gives:

$$\begin{aligned} \frac{\text{Rate}_1}{\text{Rate}_4} &= \frac{k'[\text{Pt(IV)}]_1^m[\text{R-SH}]_1^n}{k'[\text{Pt(IV)}]_4^m[\text{R-SH}]_4^n} \\ \frac{1.6 \times 10^{-4}}{8.0 \times 10^{-5}} &= \frac{k' \cdot (3.0 \times 10^{-4})^m \cdot (3.0 \times 10^{-4})^n}{k' \cdot (3.0 \times 10^{-4})^m \cdot (1.5 \times 10^{-4})^n} \\ 2.0 &= (2.0)^n \\ n &= 1 \end{aligned}$$

Therefore, the rate law for the overall reaction is:

$$\text{Rate} = k'[\text{Pt(IV)}][\text{R-SH}]$$

(Choice A) The data show that the rate depends on both [Pt(IV)] and [R-SH].

(Choice C) The overall reaction does involve two molecules of R-SH; however, this is *not* an **elementary reaction** because it involves two steps. As such, the value of n in the rate law is *unrelated* to the **stoichiometric coefficient** of R-SH in the net reaction equation and must be determined experimentally. The data show that the rate is first order with respect

to [R-SH] (ie, doubling [R-SH] doubles the rate when [Pt(IV)] is held constant). Accordingly, the [R-SH] term must have an exponent of $n = 1$, not 2.

(Choice D) Rate laws are expressed in terms of reactants, not intermediates.

Educational objective:

A rate law relates the rate of a reaction to the molar concentration of each reactant. The dependence of the rate with respect to each reactant (ie, the exponents of the rate law) must be determined experimentally by evaluating kinetic data.

Time spent:0 sec

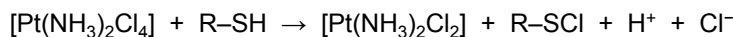
Subject:
General Chemistry

QID:403763

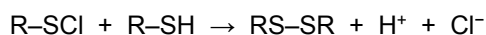
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In the first step of the mechanism, cisplatin is generated along with a transient chlorothiolate anion (R-SCI) intermediate. In the second step, the intermediate reacts with a second R-SH molecule to form the disulfide oxidation product (RS-SR). The researchers found that the activation energy for Step 1 was significantly greater than the activation energy for Step 2.

A kinetic analysis was then performed by measuring the initial rate of the reaction in six different 20.0-mL trial solutions containing different initial molar concentrations of the Pt(IV) complex and R-SH . All reactions were conducted at 25 °C in solutions with physiological pH of 7.3. The results of the analysis are given in Table 1.

Table 1 Kinetic Analysis for the Reaction of $cis\text{-}[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$ and R-SH at 25 °C in Solutions with pH 7.3

Trial	[Pt(IV)]	[R-SH]	Initial rate (M/s)
1	3.0×10^{-4}	3.0×10^{-4}	1.6×10^{-4}
2	3.0×10^{-4}	2.5×10^{-4}	1.3×10^{-4}
3	3.0×10^{-4}	2.0×10^{-4}	1.1×10^{-4}
4	3.0×10^{-4}	1.5×10^{-4}	8.0×10^{-5}
5	3.0×10^{-4}	1.0×10^{-4}	5.4×10^{-5}
6	1.5×10^{-4}	2.0×10^{-4}	5.4×10^{-5}

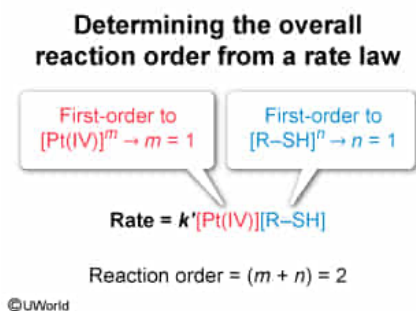
For the overall reaction to convert R-SH to the disulfide oxidation product, the overall rate constant k' was found to vary with the pH of the solution. Measured values for k' at several pH values are listed in Table 2.

Table 2 Dependence of the Overall Rate Constant k' on the Reaction Solution pH

pH	k' ($\text{M}^{-1}\text{s}^{-1}$)
2.58	8×10^{-1}
2.91	4×10^0
4.07	3×10^0
5.12	3×10^1
6.11	7×10^2
8.55	2×10^4
9.08	5×10^4
9.94	1×10^5

Based on the rate law determined from the data in Table 1, the overall order of the reaction to convert R-SH to the disulfide oxidation product is:

- ☐ A. zeroth order. [4%]
☐ B. first order. [23%]
☒ C. second order. [66%]
☐ D. third order. [4%]

Explanation:

A **rate law** relates the rate of a reaction to the **molar concentration** of each reactant and is expressed generally by the equation:

$$\text{Rate} = k[\text{A}]^m[\text{B}]^n$$

where k is a constant and m and n are the partial **reaction orders** with respect to reactants A and B. As such, the **overall order** of the reaction is given by the **sum of the partial orders**.

$$\text{Reaction order (overall)} = m + n$$

The values of m and n must be experimentally determined by comparing the reactant concentrations and the initial rates in the given reaction kinetics data. For the kinetic data given in the passage:

- **Comparing Trial 6 to Trial 3** shows that doubling $[\text{Pt(IV)}]$ while $[\text{R-SH}]$ is left unchanged causes the rate to double. As such, the reaction rate has a **first-order dependence on $[\text{Pt(IV)}]$** (ie, $[\text{Pt(IV)}]^m = 2^m = 2$, $m = 1$).
- **Comparing Trial 5 to Trial 3** shows that doubling $[\text{R-SH}]$ while $[\text{Pt(IV)}]$ is left unchanged causes the rate to double. Accordingly, the reaction rate has a **first-order dependence on $[\text{R-SH}]$** (ie, $[\text{R-SH}]^n = 2^n = 2$, $n = 1$).

As such, the rate law for the overall reaction is:

$$\text{Rate} = k'[\text{Pt(IV)}][\text{R-SH}]$$

Therefore, the overall reaction is **second order**:

$$\text{Reaction order} = (m + n) = (1 + 1) = 2$$

(Choice A) The rate of a zeroth-order reaction does not change when the reactant concentrations increase or decrease. The kinetics data show that the rate of the given reaction *does vary* as the concentrations of the reactants are changed, and therefore the reaction cannot have an order of zero.

(Choice B) The partial reaction order with respect to both $[\text{Pt(IV)}]$ and $[\text{R-SH}]$ is 1, but this is not the *overall* reaction order, which is the *sum* of the partial orders.

(Choice D) A third-order reaction would require the partial orders of the reactants to sum to 3, but this is not supported by the kinetic analysis in Table 1.

Educational objective:

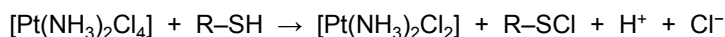
A rate law relates the rate of a reaction to the molar concentration of each reactant. The partial reaction order with respect to each reactant must be determined experimentally by evaluating kinetic data. The overall order of a reaction is the sum of the partial orders of each reactant in the rate law.

Time spent: 0 sec
 Subject:
 General Chemistry

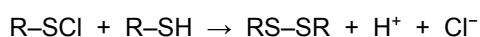
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Reaction 1 (Step 1)



Reaction 2 (Step 2)

In the first step of the mechanism, cisplatin is generated along with a transient chlorothiolate anion (R-SCI) intermediate. In the second step, the intermediate reacts with a second R-SH molecule to form the disulfide oxidation product (RS-SR). The researchers found that the activation energy for Step 1 was significantly greater than the activation energy for Step 2.

A kinetic analysis was then performed by measuring the initial rate of the reaction in six different 20.0-mL trial solutions containing different initial molar concentrations of the Pt(IV) complex and R-SH . All reactions were conducted at 25 °C in solutions with physiological pH of 7.3. The results of the analysis are given in Table 1.

Table 1 Kinetic Analysis for the Reaction of $cis\text{-[Pt(NH}_3)_2\text{Cl}_4]$ and R-SH at 25 °C in Solutions with pH 7.3

Trial	[Pt(IV)]	[R-SH]	Initial rate (M/s)
1	3.0×10^{-4}	3.0×10^{-4}	1.6×10^{-4}
2	3.0×10^{-4}	2.5×10^{-4}	1.3×10^{-4}
3	3.0×10^{-4}	2.0×10^{-4}	1.1×10^{-4}
4	3.0×10^{-4}	1.5×10^{-4}	8.0×10^{-5}
5	3.0×10^{-4}	1.0×10^{-4}	5.4×10^{-5}
6	1.5×10^{-4}	2.0×10^{-4}	5.4×10^{-5}

For the overall reaction to convert R-SH to the disulfide oxidation product, the overall rate constant k' was found to vary with the pH of the solution. Measured values for k' at several pH values are listed in Table 2.

Table 2 Dependence of the Overall Rate Constant k' on the Reaction Solution pH

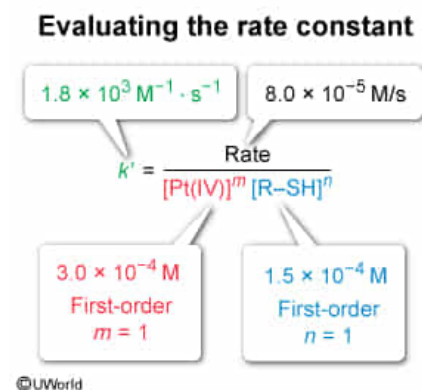
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6.11	7×10^2
8.55	2×10^4
9.08	5×10^4
9.94	1×10^5

What is the rate constant k' for the overall reaction to convert R-SH to the disulfide oxidation product at physiological pH of 7.3?

- ☐ A. $1.6 \times 10^{-4} \text{ M}^{-1}\cdot\text{s}^{-1}$ [4%]
- ☐ B. $5.3 \times 10^{-1} \text{ M}^{-1}\cdot\text{s}^{-1}$ [3%]
- ☒ C. $1.8 \times 10^3 \text{ M}^{-1}\cdot\text{s}^{-1}$ [72%]
- ☐ D. $2.6 \times 10^4 \text{ M}^{-1}\cdot\text{s}^{-1}$ [19%]

Correct

74% answered correctly

Explanation:

The general **rate law** for a reaction with two reactants (A and B) is given by the equation:

$$\text{Rate} = k[\text{A}]^m[\text{B}]^n$$

Substituting the Pt(IV) complex and R-SH reactants from the reaction in the passage into the general rate law gives:

$$\text{Rate} = k'[\text{Pt(IV)}]^m[\text{R-SH}]^n$$

The values of m and n , which give the partial **reaction order** with respect to each reactant, can be determined by comparing the initial reactant concentrations and rates from the **kinetic data** given in Table 1. The data show that doubling either [Pt(IV)] or [R-SH] while the other is left unchanged causes the reaction rate to double (eg, compare Trial 6 to Trial 3 or compare Trial 5 to Trial 3). This indicates a **first-order dependence** on both reactants:

$$[\text{Pt(IV)}]^m = 2^m = 2 \Rightarrow m = 1$$

$$[\text{R-SH}]^n = 2^n = 2 \Rightarrow n = 1$$

As a result, the rate law for the overall reaction is:

$$\text{Rate} = k'[\text{Pt(IV)}][\text{R-SH}]$$

Rearranging the rate law to solve for the **rate constant k'** yields the expression:

$$k' = \frac{\text{Rate}}{[\text{Pt(IV)}][\text{R-SH}]}$$

Substituting the values from any row of Table 1 into the expression (data from Trial 4 used below) and **solving** for k' gives:

$$k' = \frac{8.0 \times 10^{-5} \text{ M/s}}{(3.0 \times 10^{-4} \text{ M})(1.5 \times 10^{-4} \text{ M})} = \frac{8.0 \times 10^{-5} \text{ M/s}}{4.5 \times 10^{-8} \text{ M}^2} = \frac{8.0}{4.5} \times 10^{-5 - (-8)} \frac{1}{\text{M} \cdot \text{s}} = 1.8 \times 10^3 \text{ M}^{-1} \cdot \text{s}^{-1}$$

(Choice A) A value of $1.6 \times 10^{-4} \text{ M}^{-1} \cdot \text{s}^{-1}$ is found by mistaking the *rate* from Trial 1 for the rate constant.

(Choice B) A rate constant of $5.3 \times 10^{-1} \text{ M}^{-1} \cdot \text{s}^{-1}$ is calculated by assuming a rate law of:

$$\text{Rate} = k'[\text{R-SH}]$$

However, the data in Table 1 show that the rate law depends on *both* [Pt(IV)] and [R-SH].

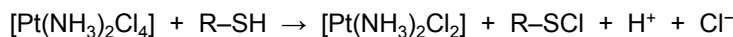
(Choice D) A rate constant of $2.6 \times 10^4 \text{ M}^{-1} \cdot \text{s}^{-1}$ is obtained by evaluating the average of the rate constants for pH 6.1 and pH 8.5 from Table 2 rather than directly calculating k' from the data in Table 1 (measured at a physiological pH of 7.3).

Educational objective:

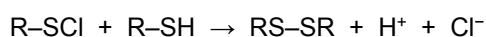
The reaction order of each reactant in a chemical reaction can be determined by comparing kinetic data at different concentrations, from which a rate law can be written. The rate constant k can then be evaluated from the rate law and the kinetic data.

Prodrugs are compounds that are metabolized into pharmacologically active substances within biological systems. The use of a prodrug can be advantageous if the prodrug compound is easier to produce, more stable, less expensive, or easier to deliver than the desired pharmaceutical.

The Pt(IV) complex $cis\text{-}[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$ has been found to act as a prodrug by reacting with the S–H bond of thiols in the cytosol of a cell to produce the Pt-based antineoplastic chemotherapy drug cisplatin ($cis\text{-}[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$) and a disulfide oxidation product. Researchers studying the reaction between $cis\text{-}[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$ and the thiolactate anion (R-SH , where $\text{R} = -\text{CH}(\text{CH}_3)\text{CO}_2^-$) as a model thiol compound proposed that the reaction proceeds in two steps according to the mechanism shown by Reactions 1 and 2.



Reaction 1 (Step 1)



Reaction 2 (Step 2)

In the first step of the mechanism, cisplatin is generated along with a transient chlorothiolate anion (R-SCI) intermediate. In the second step, the intermediate reacts with a second R-SH molecule to form the disulfide oxidation product (RS-SR). The researchers found that the activation energy for Step 1 was significantly greater than the activation energy for Step 2.

A kinetic analysis was then performed by measuring the initial rate of the reaction in six different 20.0-mL trial solutions containing different initial molar concentrations of the Pt(IV) complex and R-SH . All reactions were conducted at 25 °C in solutions with physiological pH of 7.3. The results of the analysis are given in Table 1.

Table 1 Kinetic Analysis for the Reaction of $cis\text{-}[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$ and R-SH at 25 °C in Solutions with pH 7.3

Trial	[Pt(IV)]	[R-SH]	Initial rate (M/s)
1	3.0×10^{-4}	3.0×10^{-4}	1.6×10^{-4}
2	3.0×10^{-4}	2.5×10^{-4}	1.3×10^{-4}
3	3.0×10^{-4}	2.0×10^{-4}	1.1×10^{-4}
4	3.0×10^{-4}	1.5×10^{-4}	8.0×10^{-5}
5	3.0×10^{-4}	1.0×10^{-4}	5.4×10^{-5}
6	1.5×10^{-4}	2.0×10^{-4}	5.4×10^{-5}

For the overall reaction to convert R-SH to the disulfide oxidation product, the overall rate constant k' was found to vary with the pH of the solution. Measured values for k' at several pH values are listed in Table 2.

Table 2 Dependence of the Overall Rate Constant k' on the Reaction Solution pH

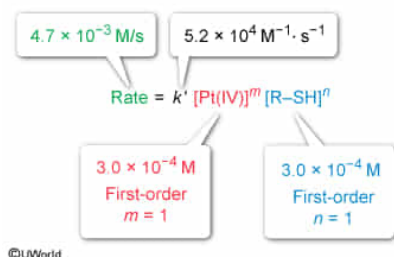
pH	k' ($\text{M}^{-1}\text{s}^{-1}$)
5.8	$2.58.8 \times 10^{-1}$
6.1	$2.91.4 \times 10^0$
6.4	$4.07.3 \times 10^0$
6.7	$5.12.3 \times 10^1$
7.0	$6.11.7 \times 10^2$
7.3	$8.55.2 \times 10^4$
7.6	$9.08.5 \times 10^4$
7.9	$9.94.1 \times 10^5$

What is the initial rate of the overall reaction at 25 °C and pH 8.5 if the initial concentration of each reactant is 3.0×10^{-4} M?

- ☐ A. 1.6×10^{-4} M/s [14%]
- ✓ ☒ B. 4.7×10^{-3} M/s [64%]
- ☐ C. 1.6×10^1 M/s [11%]
- ☐ D. 5.2×10^4 M/s [9%]

Explanation:

Evaluating the rate using the rate law



The general **rate law** for a reaction with two reactants (A and B) is given by the equation:

$$\text{Rate} = k[\text{A}]^m[\text{B}]^n$$

Accordingly, the rate law for the reaction between the Pt(IV) complex and R-SH is given by:

$$\text{Rate} = k' [\text{Pt(IV)}]^m [\text{R-SH}]^n$$

In Trial 3, the concentration of the Pt(IV) complex ($3.0 \times 10^{-4} \text{ M}$) is 2 times greater than in Trial 6 ($1.5 \times 10^{-4} \text{ M}$), and the rate is also 2 times greater in Trial 3 ($1.1 \times 10^{-4} \text{ M/s}$) than in Trial 6 ($5.4 \times 10^{-5} \text{ M/s}$). Therefore, the reaction rate is **directly proportional** to the concentration of the Pt(IV) complex, and $m = 1$ (ie, $[\text{Pt(IV)}]^m = 2^m = 2 \rightarrow m = 1$).

Similarly, in Trial 1, the concentration of R-SH is 2 times greater than in Trial 4, and the rate in Trial 1 is also 2 times greater than in Trial 4. Therefore, the reaction rate is also directly proportional to the concentration of R-SH, and $n = 1$ (ie, $[\text{R-SH}]^n = 2^n = 2 \rightarrow n = 1$). Substituting these values into the general rate law (where exponents of 1 are not explicitly shown) yields:

$$\text{Rate} = k' [\text{Pt(IV)}][\text{R-SH}]$$

According to Table 2, the measured value of k' at 25°C and pH 8.5 is $5.2 \times 10^4 \text{ M}^{-1} \cdot \text{s}^{-1}$. If the initial concentration of each reactant is $3.0 \times 10^{-4} \text{ M}$, substituting these values into the expression and **solving** for the rate gives:

$$\text{Rate} = (5.2 \times 10^4 \text{ M}^{-1} \cdot \text{s}^{-1})(3.0 \times 10^{-4} \text{ M})(3.0 \times 10^{-4} \text{ M}) = (5.2 \times 10^4 \text{ M}^{-1} \cdot \text{s}^{-1})(9.0 \times 10^{-8} \text{ M}^2)$$

$$\text{Rate} = (5.2)(9.0) \times 10^{4+(-8)} \text{ M/s} = 46.8 \times 10^{-4} \text{ M/s} \Rightarrow 4.7 \times 10^{-3} \text{ M/s}$$

(Choice A) A rate of $1.6 \times 10^{-4} \text{ M/s}$ is observed at pH 7.3 (Table 1), not at pH 8.5.

(Choice C) A rate of $1.6 \times 10^1 \text{ M/s}$ is calculated by assuming a rate law of:

$$\text{Rate} = k' [\text{R-SH}]$$

However, the data in Table 1 show that the rate law depends on *both* $[\text{Pt(IV)}]$ and $[\text{R-SH}]$.

(Choice D) A rate of $5.2 \times 10^4 \text{ M/s}$ is found by mistaking the *rate constant* for the rate.

Educational objective:

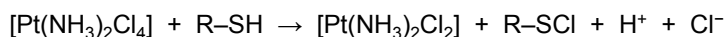
The initial rate of a chemical reaction under a given set of conditions can be calculated from the rate law if the rate constant k and initial concentrations are known.

Time spent: 0 sec
Subject:
General Chemistry

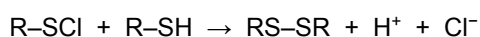
QID: 403766
System:
Thermodynamics, Kinetics & Gas Laws

Prodrugs are compounds that are metabolized into pharmacologically active substances within biological systems. The use of a prodrug can be advantageous if the prodrug compound is easier to produce, more stable, less expensive, or easier to deliver than the desired pharmaceutical.

The Pt(IV) complex $cis\text{-}[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$ has been found to act as a prodrug by reacting with the S–H bond of thiols in the cytosol of a cell to produce the Pt-based antineoplastic chemotherapy drug cisplatin ($cis\text{-}[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$) and a disulfide oxidation product. Researchers studying the reaction between $cis\text{-}[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$ and the thiolactate anion (R-SH , where $\text{R} = -\text{CH}(\text{CH}_3)\text{CO}_2^-$) as a model thiol compound proposed that the reaction proceeds in two steps according to the mechanism shown by Reactions 1 and 2.



Reaction 1 (Step 1)



Reaction 2 (Step 2)

In the first step of the mechanism, cisplatin is generated along with a transient chlorothiolate anion (R-SCl) intermediate. In the second step, the intermediate reacts with a second R-SH molecule to form the disulfide oxidation product (RS-SR). The researchers found that the activation energy for Step 1 was significantly greater than the activation energy for Step 2.

A kinetic analysis was then performed by measuring the initial rate of the reaction in six different 20.0-mL trial solutions containing different initial molar concentrations of the Pt(IV) complex and R-SH . All reactions were conducted at 25 °C in solutions with physiological pH of 7.3. The results of the analysis are given in Table 1.

Table 1 Kinetic Analysis for the Reaction of $cis\text{-}[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$ and R-SH at 25 °C in Solutions with pH 7.3

Trial	[Pt(IV)]	[R-SH]	Initial rate (M/s)
1	3.0×10^{-4}	3.0×10^{-4}	1.6×10^{-4}
2	3.0×10^{-4}	2.5×10^{-4}	1.3×10^{-4}
3	3.0×10^{-4}	2.0×10^{-4}	1.1×10^{-4}
4	3.0×10^{-4}	1.5×10^{-4}	8.0×10^{-5}
5	3.0×10^{-4}	1.0×10^{-4}	5.4×10^{-5}
6	1.5×10^{-4}	2.0×10^{-4}	5.4×10^{-5}

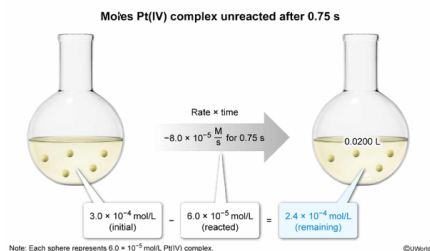
For the overall reaction to convert R-SH to the disulfide oxidation product, the overall rate constant k' was found to vary with the pH of the solution. Measured values for k' at several pH values are listed in Table 2.

Table 2 Dependence of the Overall Rate Constant k' on the Reaction Solution pH

pH	k' ($\text{M}^{-1}\text{s}^{-1}$)
2.58	8×10^{-1}
2.91	4×10^0
4.07	3×10^0
5.12	3×10^1
6.11	7×10^2
8.55	2×10^4
9.08	5×10^4
9.94	1×10^5

During the reaction performed in Trial 4 of the experiment in Table 1, what is the approximate molar concentration (molarity) of the unreacted Pt(IV) complex remaining in the solution after 0.75 s? (Note: Assume that the average rate of the reaction during the first 0.75 s is approximately equal to the initial rate.)

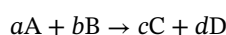
- ☐ A. 4.8×10^{-6} M [5%]
☐ B. 6.0×10^{-5} M [27%]
☒ C. 2.4×10^{-4} M [56%]
☐ D. 3.0×10^{-4} M [9%]

Explanation:

Reaction rates indicate how quickly a chemical conversion happens by measuring the change in the concentration of the reactants or products per unit of time. An **average rate** measures the change in concentration $\Delta[X]$ over a time interval Δt .

$$\text{Average rate} = \frac{\text{Change in concentration of X}}{\text{Change in time}} = \frac{[X]_{\text{final}} - [X]_{\text{initial}}}{t_{\text{final}} - t_{\text{initial}}} = \frac{\Delta[X]}{\Delta t}$$

For a general reaction:



The rates for the consumption or production of individual species are related to each other and to the overall reaction rate by the **stoichiometric coefficients** of the reaction:

$$\text{Reaction rate} = -\frac{1}{a} \cdot \frac{\Delta[A]}{\Delta t} = -\frac{1}{b} \cdot \frac{\Delta[B]}{\Delta t} = \frac{1}{c} \cdot \frac{\Delta[C]}{\Delta t} = \frac{1}{d} \cdot \frac{\Delta[D]}{\Delta t}$$

The reactants include a negative sign because the concentrations *decrease* as the reaction proceeds, whereas the concentrations of products *increase* and have positive rate expressions.

Based on the net reaction from the mechanism given by Reactions 1 and 2:



the overall rate of the reaction is related to the rate of the consumption of the Pt(IV) complex according to the expression:

$$\text{Reaction rate} = -\frac{1}{1} \cdot \frac{\Delta[\text{Pt(IV)}]}{\Delta t}$$

Substituting the measured rate of the reaction for Trial 4 into the equation gives the initial rate at which the [Pt(IV)] decreases as it is consumed in the reaction:

$$8.0 \times 10^{-5} \text{ M/s} = -\frac{1}{1} \cdot \frac{\Delta[\text{Pt(IV)}]}{\Delta t}$$

$$\frac{\Delta[\text{Pt(IV)}]}{\Delta t} = -8.0 \times 10^{-5} \text{ M/s}$$

If the reaction proceeds at this rate for 0.75 s, the change in the **molar concentration** (ie, M = mol/L) of the Pt(IV) complex in the solution is:

$$\Delta[\text{Pt(IV)}] = \left(-8.0 \times 10^{-5} \frac{\text{M}}{\text{s}}\right) \times 0.75 \text{ s} = -6.0 \times 10^{-5} \text{ M}$$

The difference between the initial concentration and the change in the concentration due to the number of moles that reacted gives the concentration of unreacted Pt(IV) complex remaining in the solution.

$$[\text{Pt(IV)}]_{\text{Initial}} - [\text{Pt(IV)}]_{\text{Reacted}} = [\text{Pt(IV)}]_{\text{Remaining}}$$

$$3.0 \times 10^{-4} \text{ M (initial)} - 6.0 \times 10^{-5} \text{ M (reacted)} = 2.4 \times 10^{-4} \text{ M (remaining)}$$

(Choice A) A value of $4.8 \times 10^{-6} \text{ M}$ results from mistaking the number of *moles* of unreacted Pt(IV) complex for its *molarity* (ie, molar concentration):

$$0.0200 \text{ L solution} \times \frac{2.4 \times 10^{-4} \text{ moles Pt(IV)}}{\text{L solution}} = 4.8 \times 10^{-6} \text{ moles Pt(IV) unreacted}$$

(Choice B) A value of 6.0×10^{-5} M is the *change* in the concentration of the Pt(IV) complex due to the number of moles of the complex that react and are consumed by the reaction in the first 0.75 s, but this is not the concentration of the Pt(IV) complex that remains unreacted.

(Choice D) A value of 3.0×10^{-4} M is the concentration of Pt(IV) complex *initially present* before the reaction begins.

Educational objective:

A reaction rate measures the change in the concentration of the reactants or products per unit of time. Rates for individual species are related to the overall rate by the stoichiometric coefficients of the reaction. The changes can be expressed in terms of moles by multiplying the molar concentrations by the solution volume.

Time spent:0 sec

Subject:
General Chemistry

QID:403767

System:
Thermodynamics, Kinetics & Gas Laws